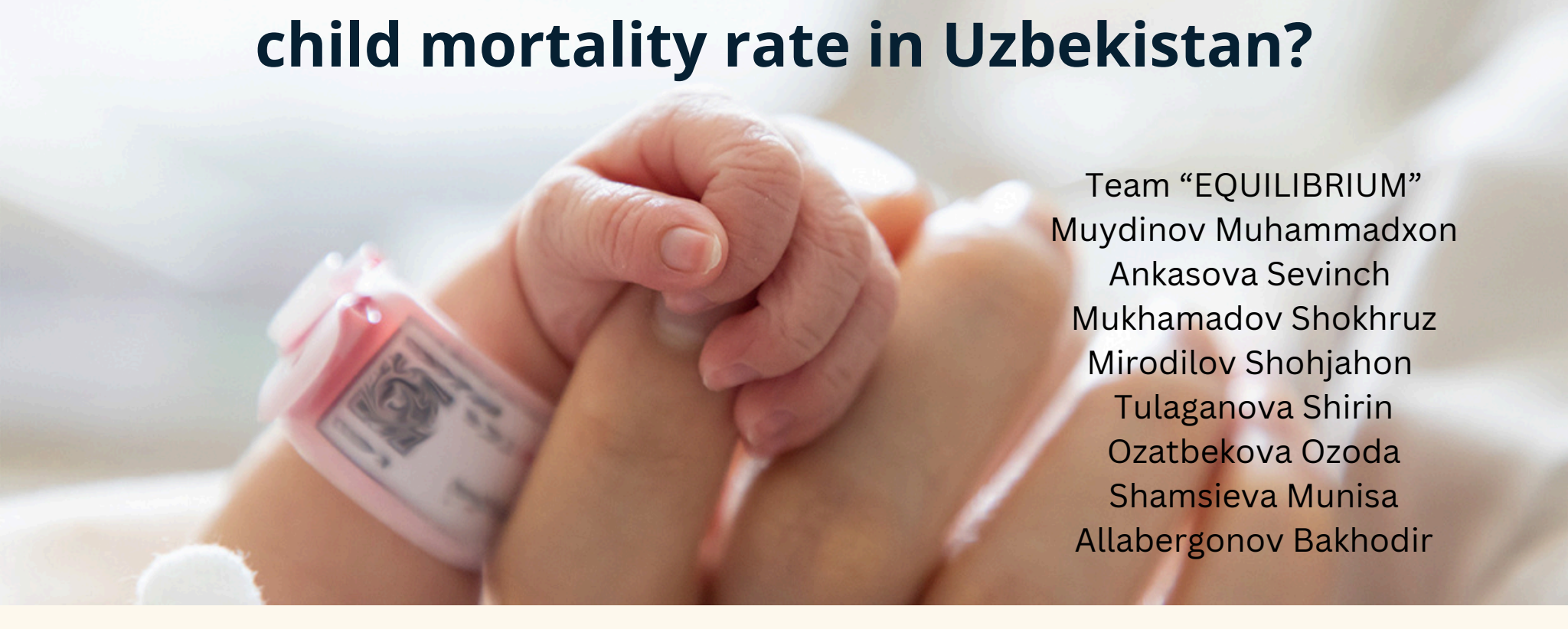


# How do characteristics of urbanisation influence child mortality rate in Uzbekistan?




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## Introduction

This research is significant and innovative because it addresses a critical gap in the existing literature by focusing on Uzbekistan, employs advanced methodologies for robust analysis, and considers the unique socio-economic and cultural context of Central Asia.

By doing so, not only enhances academic understanding of urbanization and child mortality but also provides valuable insights for policymakers and practitioners aiming to improve child health outcomes in rapidly urbanizing environments.

## Previous studies and lessons to be learnt

Urbanization and Child Mortality – Evidence from the Demographic and Health Surveys (G.Fink & K. Hill, by Harvard School of Public Health, June 2013)

The study examines the impact of urbanisation on child mortality using data from 37 low- and middle-income countries over two decades. It finds that, despite rapid urbanization, rural areas still account for the majority of child deaths.

Urban areas, huge cities, show significantly lower child mortality rates compared to rural and smaller urban areas. This is attributed to better healthcare access, higher education levels, and more investments in large cities. However, the mortality gap between rural and urban children remains substantial.

## Limitations

- Data Consistency: The study periods are based on available survey data, not strictly on a decade-long interval, potentially affecting the reported changes.

- Causal Determinants: The study does not delve into the specific causal factors behind the observed mortality trends, beyond changes in residential composition.

## Recommendations

- Further Research: More research is needed to understand the socioeconomic and health system changes contributing to the mortality improvements, particularly in large urban areas.

- Policy Implementation: Developing and implementing effective public health policies and interventions is essential to leverage urbanization benefits for child health outcomes.

## Hypothesis

**Null Hypothesis (H0):** Urbanisation does not significantly improve economic opportunities and does not reduce child mortality.

**Alternative Hypothesis (H1):** Urbanisation significantly improves economic opportunities, which in turn reduces child mortality

**Data sources:**

- International Monetary Fund (IMF)
- National Statistical Committee of the Republic of Uzbekistan
- United Nations (UN) Databases
- World Bank
- Other Relevant Publications and Reports

## Methodology

We used country level data on Uzbekistan (from the years 2013-2023).
The secondary data used in this research are from the The World Bank Data Base.
The independent variables that characterise urbanisation are chosen based on the literature review studied.
Employed in this study is the Multiple Regression model which acts as the analytical technique.

The econometric version of the model is:

**MORTt = a + a1ELECt + a2INTRt + a3SANIt + a4HEALt + a5UNEM + mt**

Lack of nutrition

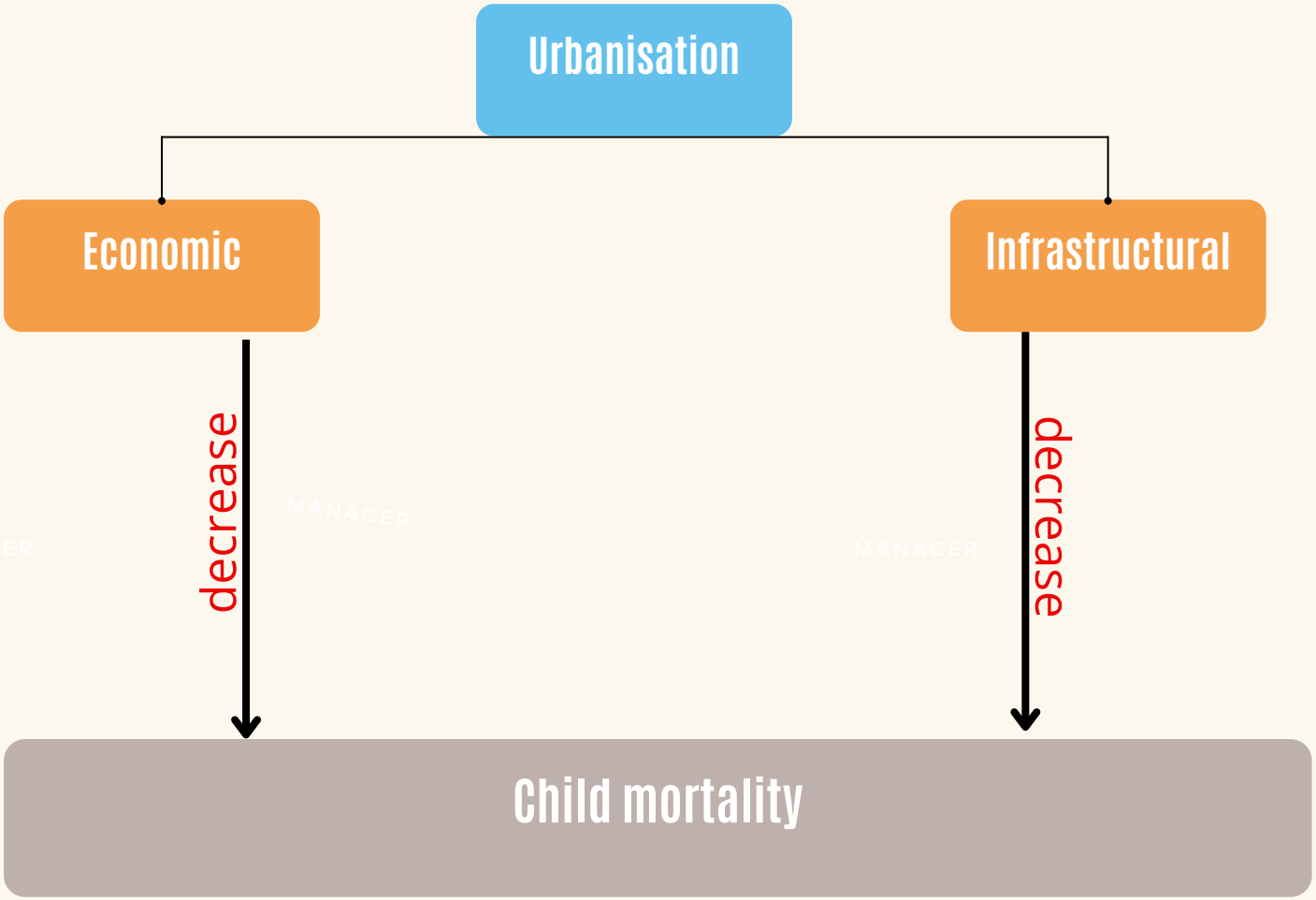
Bad access to hospitalization

# Why children die?

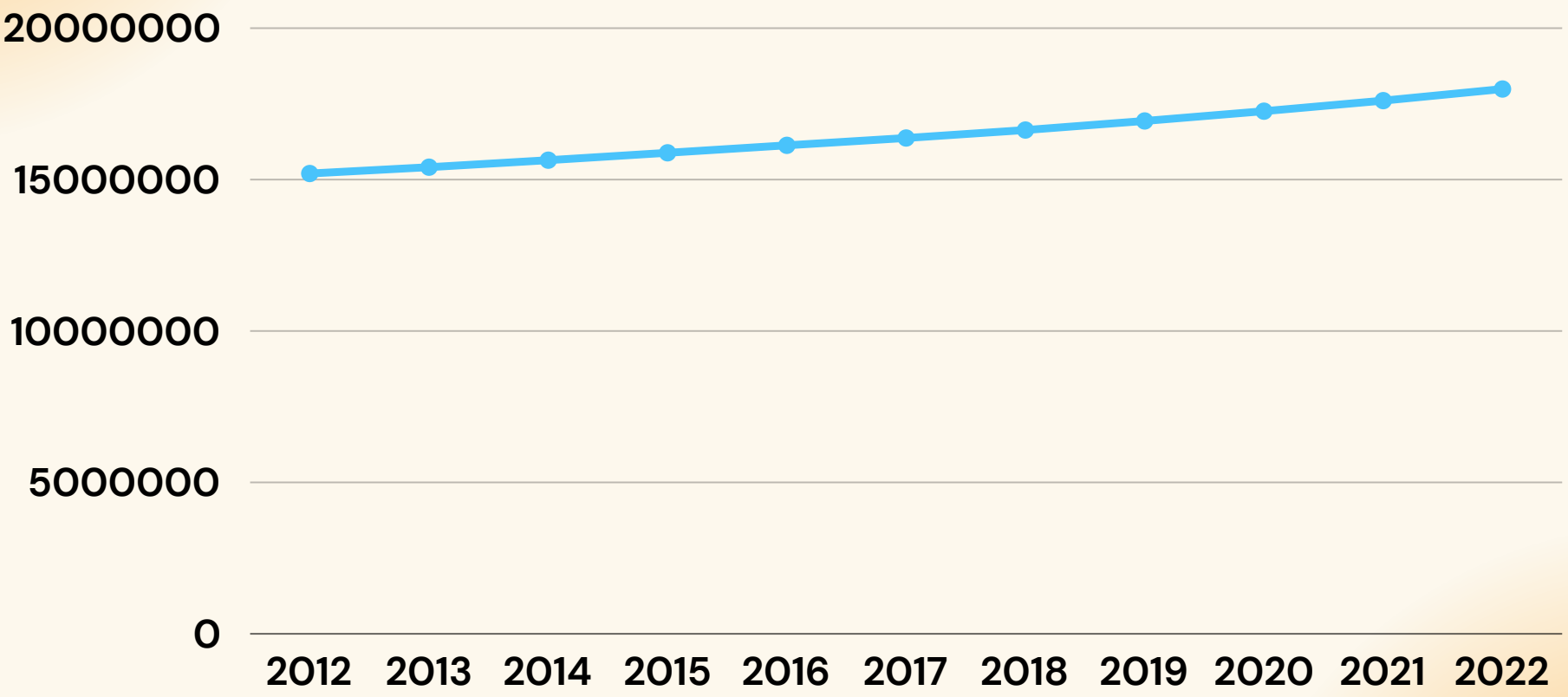
Poverty

Bad sanitation services

# CHILD MORTALITY

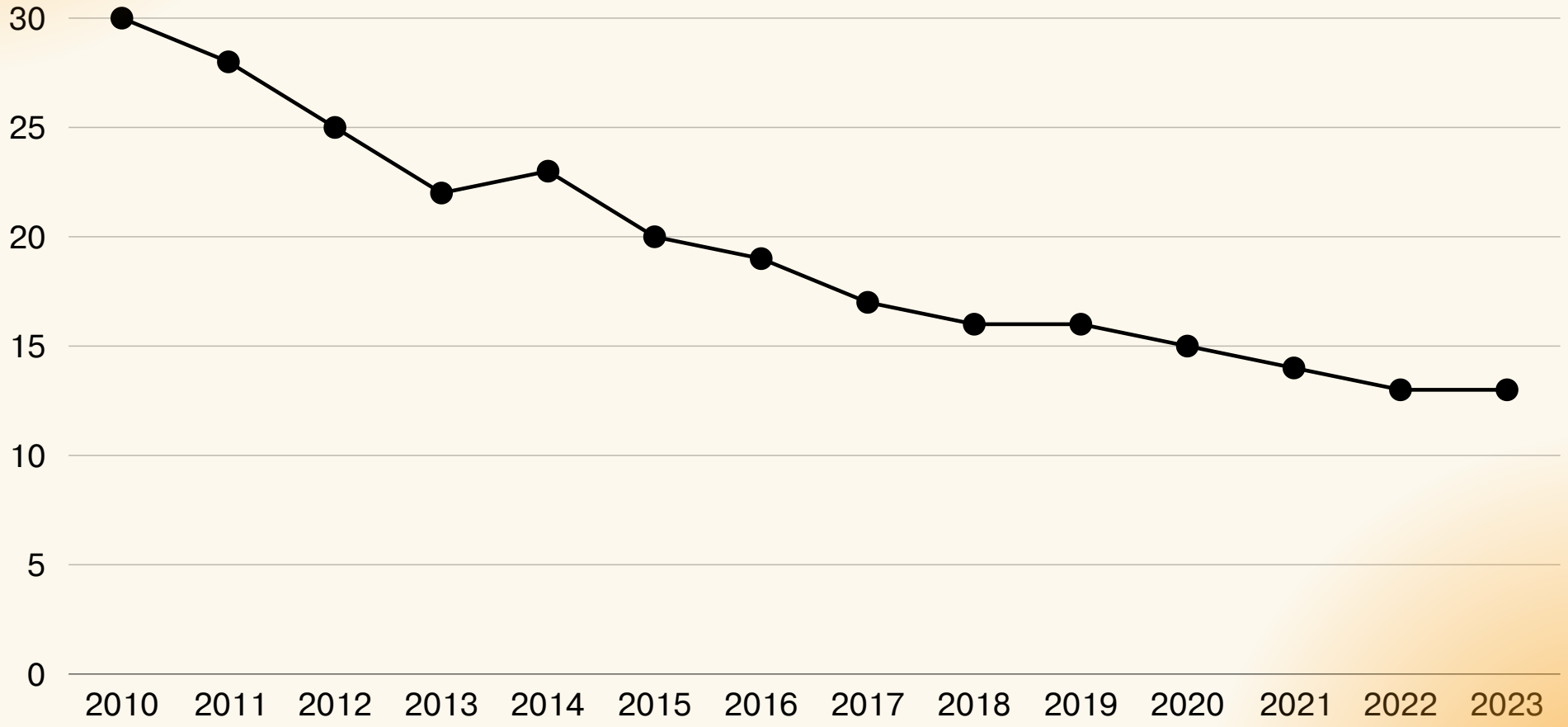


# Urban population of Uzbekistan

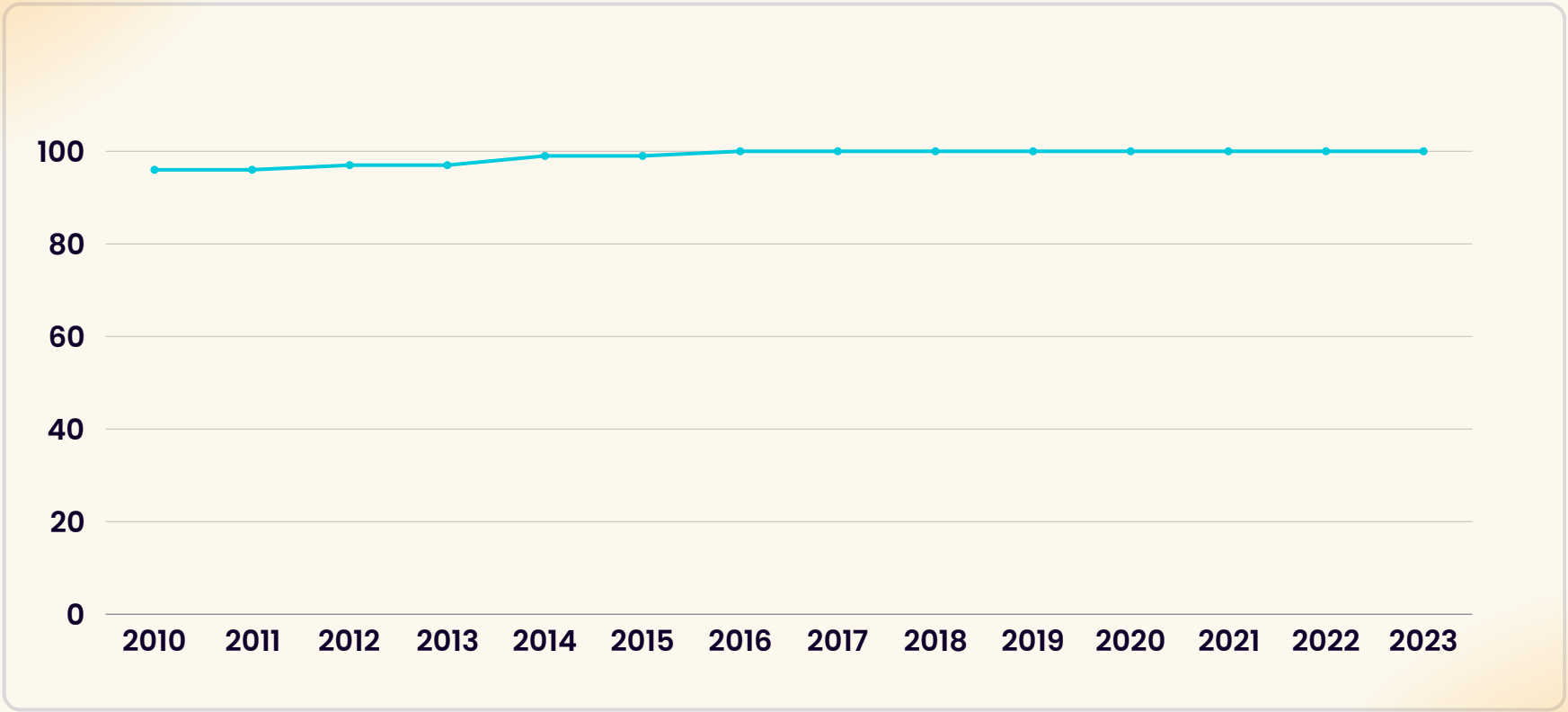



Period 2012-2022

# Infant mortality(in promille)



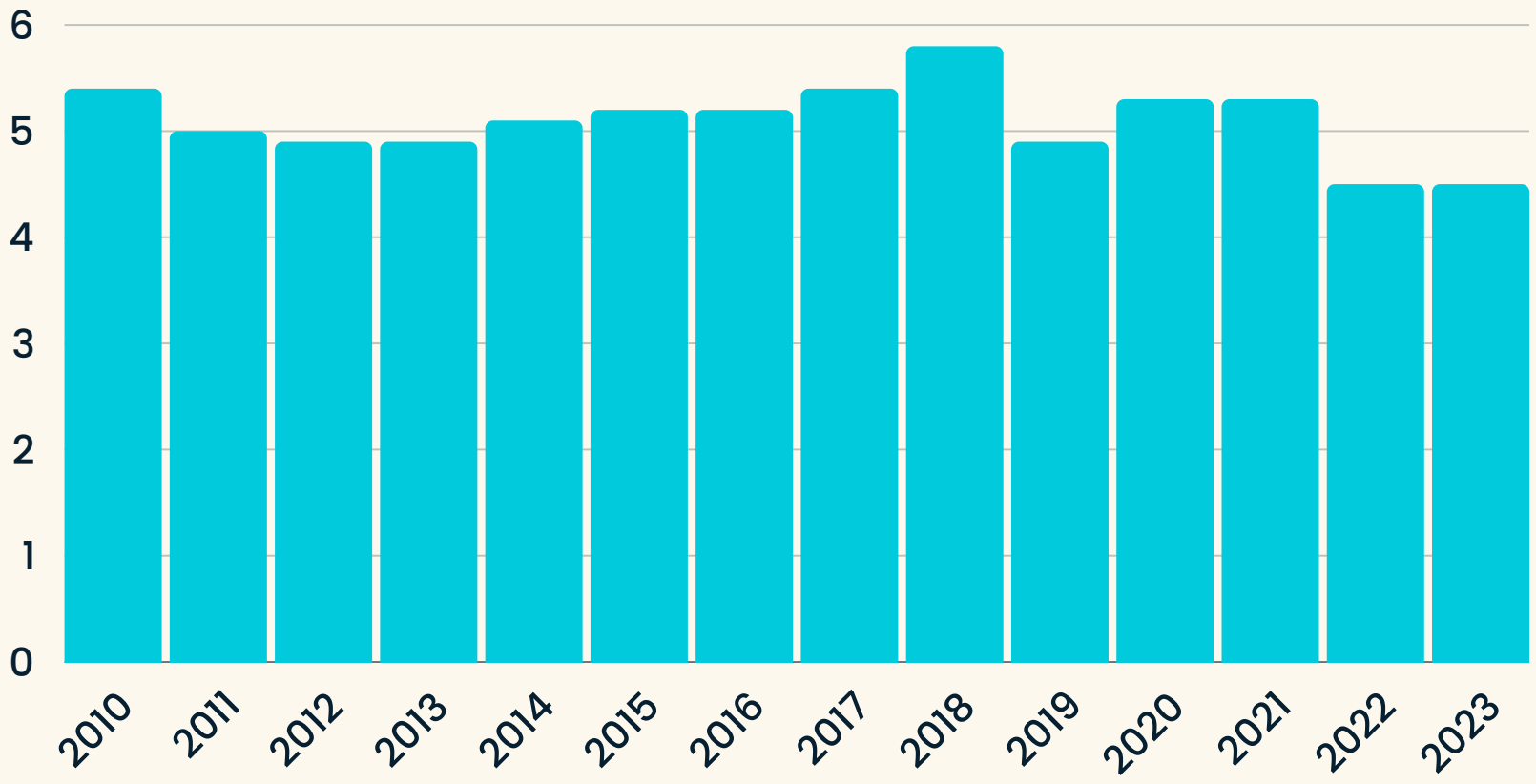
# Access to electricity in Uzbekistan(in percentages)



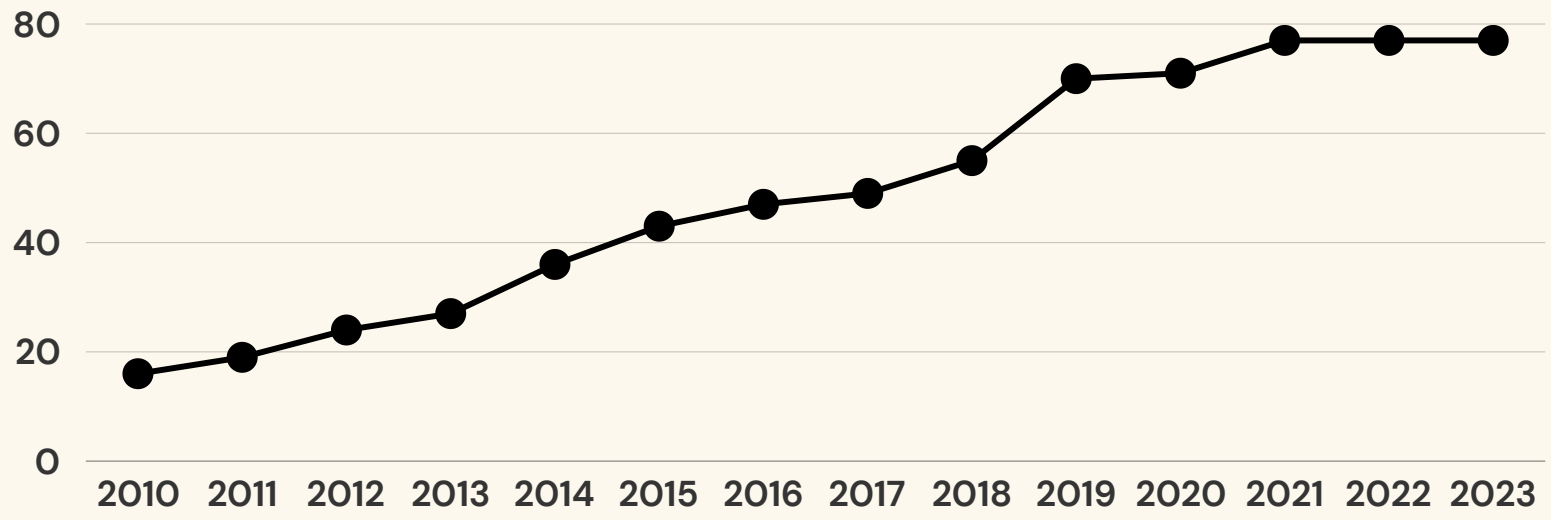

Period 2010-2023

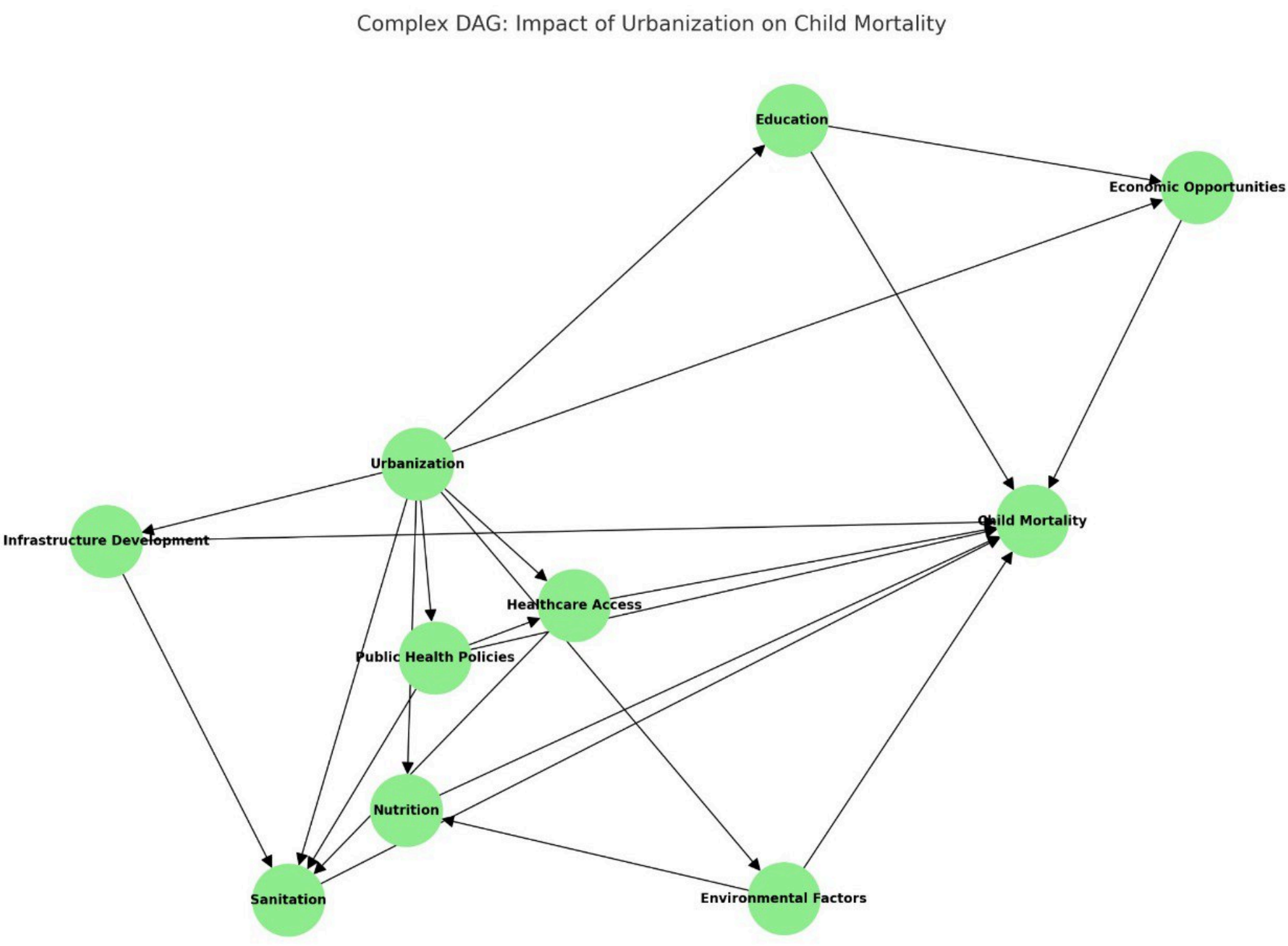
# Unemployment rate in Uzbekistan (in percentages)

Period 2010-2023



# Rate of individuals using the internet in Uzbekistan (2010-2023)




Complex DAG: Impact of Urbanization on Child Mortality
Education
Economic Opportunities
Urbanization
Infrastructure Development
Child Mortality
Healthcare Access
Public Health Policies
Nutrition
Sanitation
Environmental Factors

**Recent findings about the various effects to child mortality included in DAG model**

Healthcare Access - (Bocquier et al., 2011; Van de Poel et al., 2009)

Sanitation and Infrastructure - (Prüss-Ustün et al., 2014)

Education and Economic Opportunities - (Cleland & Van Ginneken, 1988; Hobcraft, 1993)

Environmental Factors - (World Health Organization, 2016)

Nutrition - (Popkin, 2001)

Public Health Policies - (Ezeh et al., 2017)

SUMMARY OUTPUT

| *Regression Statistics* | |
|---|---|
| Multiple R | 0.999007578 |
| R Square | 0.998016142 |
| Adjusted R Square | 0.996776231 |
| Standard Error | 0.015345861 |
| Observations | 14 |

ANOVA

| | *df* | *SS* | *MS* | *F* | *S* |
|---|---|---|---|---|---|
| Regression | 5 | 0.947762341 | 0.189552468 | 804.9092848 | |
| Residual | 8 | 0.001883964 | 0.000235495 | | |
| Total | 13 | 0.949646304 | | | |

| | *Coefficients* | *Standard Error* | *t Stat* | *P-value* | |
|---|---|---|---|---|---|
| Intercept | 347.9035841 | 70.92617307 | 4.905150934 | | 0.001186028 |
| Access to electricty | -35.26305309 | 14.00549134 | -2.517801927 | | 0.035931583 |
| Individuals using the internet | -0.149656532 | 0.062822391 | -2.382216408 | | 0.044391845 |
| People using safely managed sanitatior | -42.33488872 | 11.34189323 | -3.732612171 | | 0.005766104 |
| Health expenditure | -0.038670736 | 0.050568689 | -0.764717006 | | 0.466396412 |
| Unemployment | 0.147585771 | 0.080722672 | 1.828306322 | | 0.10490898 |

**y=-35.263x1-0.149x2-42.334x3+347.9036**

y - infant mortality

x1 - access to electricity

x2 - individuals using internet

x3 - people using safely managed sanitation

## Literature review

1. "Infant and Child Mortality in India in the Last Two Decades: A Geospatial Analysis" Abhishek Singh, Praveen Kumar Pathak, Rajesh Kumar Chauhan, William Pan
2. "A Comprehensive Analysis on Child Mortality and Its Determinants in Bangladesh Using Frailty Models" Jahidur Rahman Khan, Nabil Awan
3. "Reducing Infant and Child Mortality: Assessing the Social Inclusiveness of Child Health Care Policies and Programmes in Three States of India" Madhumita Bango, Soumitra Ghosh
4. "Urbanization and Child Health in Resource Poor Settings with Special Reference to Under-Five Mortality in Africa" Michel Garenne

# THANK YOU FOR ATTENTION!